THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · MEDIUM LEVEL UNIT TEST SET 2

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. An equilateral triangle of side 12 cm and a square have the same perimeter. The side of the square (cm) is: 1

(a) 6 **(b)** 12 **(c)** 15 **(d)** 9

2. After spending $\frac{3}{8}$ of her money, Neha has ₹100 left. What fraction is left? 1

(a) $\frac{5}{9}$ **(b)** $2\frac{2}{3}$ **(c)** $\frac{5}{8}$ **(d)** $\frac{3}{8}$

3. A circle has diameter 24 cm. A concentric circle has radius 7 cm. The width of the ring (cm) is: 1

(a) 6 **(b)** 7 **(c)** 4 **(d)** 5

4. A square is rotated about its centre through 30°. The smallest further turn (degrees) after which it again coincides with itself is: 1

(a) 60 **(b)** 70 **(c)** 150 **(d)** 50

5. The difference between the highest and lowest of -8, 5, -3, 9 is: 1

(a) 18 **(b)** 16 **(c)** 17 **(d)** 19

6. A wire 40 cm long is bent into a rectangle 14 cm long. If it is bent into a square, its area (cm^2) is: 1

(a) 104 **(b)** 100 **(c)** 101 **(d)** 99

7. Which fraction lies between $\frac{1}{4}$ and $\frac{1}{2}$? 1

(a) $\frac{2}{2}$ **(b)** $\frac{1}{5}$ **(c)** $\frac{2}{4}$ **(d)** $\frac{3}{8}$

8. A square has perimeter 32 cm. Its diagonal splits it into two triangles each with two sides of length (cm): 1

(a) 7 **(b)** 24 **(c)** 9 **(d)** 8

9. The sum of the number of lines of symmetry of a square and a regular hexagon is: 1

(a) 10 **(b)** 12 **(c)** 11 **(d)** 9

10. A mountaineer is at 120 m above sea level and descends 150 m. His position (m) is: 1

(a) -30 **(b)** -28 **(c)** -32 **(d)** -20

11. A path of width 2 m is built inside a square garden of side 20 m along its boundary. The area of the remaining garden (m^2) is: 1

(a) 256 **(b)** 260 **(c)** 252 **(d)** 264

12. The numerator of $3\frac{2}{5}$ written as an improper fraction is: 1

(a) 16 **(b)** 18 **(c)** 17 **(d)** 22

13. Four equal circles of radius 3 cm are drawn touching in a row. The length of the row (cm) is: 1

(a) 24 **(b)** 36 **(c)** 27 **(d)** 21

14. How many of the letters in A, B, C, D, E have a line of symmetry? 1

(a) 5 **(b)** 7 **(c)** 6 **(d)** 4

15. Which pair gives the sum 5? 1

(a) -5 and 12 **(b)** 7 and 12 **(c)** -7 and 15 **(d)** -7 and 12

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the cost of paving a rectangular courtyard 12 m by 8 m at ₹75 per m^2. 2

17. A number increased by $\frac{1}{2}$ of itself gives 18. Find the number. 2

18. In rectangle ABCD, $\angle AOB = 40^\circ$ at the point O where the diagonals meet. Find $\angle OAB$ and $\angle AOD$. 2

19. Find the total number of lines of symmetry of a square and a regular octagon together. 2

20. Subtract 9 from (-6) and then subtract (-6) from 9. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Three circles of radius 3 cm touch each other in a row. Find the distance between the centres of the first and third circles, and the total length of the row. 3

22. Write the order of rotational symmetry and number of lines of symmetry for the letters H, A, T. Which have more lines than the order? 3

23. A bank account has ₹300. Withdrawals of ₹200 and ₹250 are made, then ₹120 is deposited. Show the balance as an integer after each step and the final balance. 3

24. A rectangular sheet 50 cm by 35 cm is cut into squares of side 10 cm. How many squares are obtained and what area is left over? 3

OR

A bottle holds 6 litres. Its $\frac{2}{3}$ part is filled. How many litres more are needed to fill it? What fraction is empty? 3

25. A rectangle has length 14 cm and its perimeter is 40 cm. Find its breadth and say whether it can be a square. 3

OR

A clock face has 12 equally spaced marks. State the number of lines of symmetry (through marks) and the angle between neighbouring marks. If the minute hand moves from mark 0 to mark 5, through what angle has it turned? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A frog is on the number line at -3. It jumps 8 units to the right, then 5 units to the left, then 9 units to the left. Find its final position, and the distance from 0. 5

OR

A square of side 20 cm has a square of side 3 cm cut from each of its four corners. Find the area of the remaining figure and its perimeter. 5

27. A tank is $\frac{1}{4}$ full. After 20 litres are added, it becomes $\frac{3}{4}$ full. Find the capacity of the tank. 5

OR

A gardener draws two concentric circles of radii 5 m and 8 m. Find the width of the path between them, and the diameters of the outer and inner circles. 5

END OF QUESTION PAPER **Total: 50 Marks**

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